

ORIGINAL ARTICLE

XGBoost-based risk prediction model for massive vehicle recalls using consumer complaints

Yi-Na Li^{1,2} | Ming Jiang² | Likun Wang² | Jiuchang Wei¹ 

¹School of Public Affairs, University of Science and Technology of China, Hefei, People's Republic of China

²School of Management, University of Science and Technology of China, Hefei, People's Republic of China

Correspondence

Jiuchang Wei, School of Public Affairs, University of Science and Technology of China, Hefei, 230026, People's Republic of China.
Email: weijc@ustc.edu.cn

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Abstract

This study employed the XGBoost model to conduct an in-depth analysis of consumer complaints and identified the key risk factors predicting vehicle recalls in the US market, providing valuable proactive risk management support for automakers and regulatory agencies. We leveraged the extensive data resources from National Highway Traffic Safety Administration to construct high-precision recall risk prediction models to predict the risk of recall. The models exhibited exceptional performance across different time windows, particularly maintaining a high level of area under the curve values over a prediction timespan of up to 18 months, demonstrating their predictive accuracy and stability. Our study contributes to risk management theory by addressing the challenges of integrating consumer complaints into predictive models for vehicle recall risk. While prior research has primarily focused on text mining of complaint content, our work systematically incorporates structured complaint data and recall records to enhance predictive accuracy. Also, our research distinguishes the indicators for the initial recall after launch to the market and the indicators for subsequent recalls, and bridges a critical gap in recall risk prediction at different stages of a vehicle's life cycle.

KEYWORDS

consumer complaints, massive vehicle recall, risk prediction, XGBoost model

1 | INTRODUCTION

Vehicle defects pose critical concerns for automakers, consumers, and government regulators. The safety-related defects endanger drivers and passengers and may lead to fatal traffic accidents and great public threats, such as individual injury and even fatality. When a defect is identified as a design or manufacturing flaw, automakers must recall all affected vehicles. A recall sometimes involves millions of cars and trucks. A prominent example is Toyota's recall crisis in 2009–2010, when the automaker, renowned for its quality and reliability, recalled millions of vehicles due to unintended acceleration caused by mechanical sticking of the accelerator pedal. As the National Highway Traffic Safety Administration (NHTSA) reported, nearly 35 million vehicles were recalled in the United States in 2023 (https://www.nhtsa.gov/sites/nhtsa.gov/files/2024-03/NHTSA-2023-Annual-Recalls-Report_0.pdf) due to defects in car seats, tires, vehicles, and other equipment. The recalls can tarnish the automakers' reputation of product reliability and result in market share reduction

(Govindaraj et al., 2004; Chu et al., 2005; Marcus, 1989; Wu & Huang, 2015), the decline in stock market value (Davidson & Worrell, 1992; Y. Chen & Nguyen, 2013; J. Z. Ni et al., 2014), and even bankruptcy (Eilert et al., 2017; Shao et al., 2024).

Automakers have invested a great effort in product quality control. They collect a wide range of data for their failure mode and effects analysis (H. C. Liu et al., 2013). For example, in the stage of manufacturing and assembly, automakers track the material and composition from suppliers' synchronized component information, use sensors to monitor the real-time process of assembly lines, and conduct comprehensive end-of-line testing to examine functional anomalies. In the after-sale stage, they gather diagnostics data from the software logs, dealer service reports, warranty claims, and regulators to detect potential defects. These established procedures across the production lifecycle aim to ensure rigorous quality control. Besides these procedures, the stakeholders' input is addressed as pivotal for risk-based decision-making (Morgan et al., 2022). Consumers' feedback, especially their complaints, provides valuable early warning signals of recur-

ring issues in real-world usage and possibly enables engineers to detect defects.

Consumer complaints are consumers' expressions of dissatisfaction and disappointment (Halstead, 2002; McGraw et al., 2015), either directly to manufacturers or indirectly on third-party platforms. However, the complaints need to be rigorously evaluated and corroborated with investigated evidence before it is used to predict defects. First, many complaints can arise from a variety of factors unrelated to product flaws, such as users' errors, misunderstanding of normal functions (e.g., an air conditioner not cooling due to an incorrect mode setting), or subjective dissatisfactions (e.g., ride discomfort or design style dislike). It is labor-intensive to eliminate the flaws unrelated information effectively through further inquiries and the confirmation of users' accountability. Second, individual differences (e.g., religious beliefs, conservative personality traits, impulsiveness, and self-control trait) influence the intention to complain (Swimberghe et al., 2009; Jung et al., 2017; Sharma et al., 2010), and how humorous (vs. serious) complaints express influences the likelihood to receive remedies and sympathy (McGraw et al., 2015). The research suggests that consumer group features and situational factors cause the bias in complaint likelihood and feedback reception. Public platforms also cause bias. They tend to amplify negative feedback—often skewing perceptions, especially for widely sold products—simply tallying complaints is insufficient and may misrepresent the true risk.

The challenges have historically limited the effective use of complaint data for risk management. NHTSA, as a regulatory agency, provides a wealth of consumer complaint and recall data on vehicle safety, which contain rich information that can be used to analyze dissatisfaction (Lapr   & Tsikriktsis, 2006; Harris & Ogbonna, 2010; Knox & Van Oest, 2014), predict opinions (Hu & Liu, 2004), or categorize product issues into potential actionable groups (Lu, 1998; Abrahams et al., 2013). However, numerous instances show that automakers ignored consumer complaints about defective products, ultimately leading to significant losses. For example, General Motors was aware of its ignition switch problem through customer complaints months before the actual recall campaign. Similarly, hundreds of reports submitted to NHTSA or the manufacturer were disregarded before Toyota's unintended acceleration crisis in 2009–2011, failing to initiate a recall (Heller & Darling, 2012). It requires a swift analysis to identify recall risk from consumer complaints and provide actionable insights for automakers to mitigate losses from vehicle defects (Law et al., 2017; Abrahams et al., 2015).

The adoption of user complaints in predicting defects can effectively enhance automakers' product risk management (Abrahams et al., 2012). First, different from the defect elimination measures reactive to the well-defined threshold and professional diagnosis, consumer complaints integrate the post-sale firsthand user information into the risk detection and identification process. The management of complaints complements the closed loop in quality control systems (Chang

& Chin, 2011; Kamada &   ry, 2020; Orsingher et al., 2010; Yilmaz et al., 2016). More crucially, it can help to reveal the undetected defect before fatal accidents or personal injuries occur. The preventive detections and risk assessment can indicate the priority for the automaker decision-makers to prepare for remedy before crisis, rather than take hazard-based reactive actions. Overall, our research provides a method to effectively engage in consumers' feedback in automakers' risk-based decision-making.

In recent years, several studies have employed various methods to predict vehicle recall events. A text mining based method (Zhang et al., 2015) utilizes user discussions on online platforms such as social media and forums as a valuable resource for predicting vehicle recalls. The RECall Incident PrEdiction (RECIPE) method (Chakraborty & Gupta, 2023) further validates the effectiveness of data-driven approaches in recall prediction by analyzing user complaints on automotive forums through a combination of graph convolutional networks (GCN) and BERT (Bidirectional Encoder Representations from Transformers) models. Other research has explored factors influencing the timing of vehicle recalls from a problem-solving perspective and proposed methods based on event history analysis (J. Ni & Huang, 2018). Two research gaps remain in vehicle recall risk prediction. First, few studies have investigated the factors influencing a vehicle model's recall risk at different stages of its life cycle. Specifically, the factors affecting a vehicle model's recall risk when it is first launched may differ from those when it is subsequently recalled. The initially launched vehicle may limit real-world trials and have production variation defects, and the later recalls may involve increasing user behavior, environmental exposure, and material degradation issues. Second, most research has focused on text mining of user complaint content while neglecting the in-depth exploration of structured data features within user complaints.

This study employs the advanced XGBoost model to predict vehicle recall risk in the US market in advance. First, the novelty of this research lies in its comprehensive examination of recall risks across different stages of a vehicle's life cycle. Unlike prior studies that assume homogeneity between stages of a vehicle's life cycle, our work systematically analyzes the predictive factors at multiple stages and discriminates the recall risk for vehicle models after their initial launch and the subsequent recall risk after the first recall. Second, this research extends the scope of defect information sources and comprehensively utilizes consumer complaints, recall records, and other publicly available information from NHTSA to extract key features influencing recall risks. We adopted the XGBoost model to predict the risk of defects. The model is excellent in handling complex non-linear relationships. Therefore, it is capable to distinguish the nuanced difference between the defect risk for the first recall after launch and that for subsequent recalls. It also has great capability in generalization and interpretability, and can offer practical guidance for regulatory agencies and manufacturers. Third, by comparing the predictive perfor-

mance of the model under different time windows, this study provides a scientific basis for designing highly proactive and resource-efficient recall management strategies, thereby enhancing the automakers' forward-looking control of safety issues.

2 | PREDICTION OF LARGE-SCALE DEFECTIVE PRODUCTS AND INFLUENCING FACTORS

Scholars have developed numerous models to identify product defects. Online product reviews offer rich information for discovering defects (Goldberg & Abrahams, 2018). The reviews can be classified into positive and negative categories based on their sentiment orientation (Archak et al., 2011). Among the negative categories, a set of vocabularies was identified for detecting defects and prioritizing defect severity (Abrahams et al., 2012). This vocabulary set was further refined, improved, and applied to the automotive industry (Zhang et al., 2015), toy industry (Winkler et al., 2016), consumer goods industry (Abrahams et al., 2015), and maternal and infant care industry (Mummalaneni et al., 2018).

In addition, scholars have employed machine learning methods to develop text classifiers for identifying product defects and potential issues. For instance, Lo (2008) used text mining and support vector machine (SVM) to automatically classify customer information, achieving an accuracy rate of over 83% in identifying defect information. Zhang et al. (2015) later proposed a supervised classification method to predict product recalls by analyzing discussions from automotive forums. Building on this work, Law et al. (2017) identify performance defects in online dishwasher reviews. Then, research advances text mining to extract, categorize, and bin numerical expressions contained in text documents to support product quality management (Gruss et al., 2018). Other studies have further refined these techniques by integrating contextual features such as "review-response" interactions (Y. Liu et al., 2018). Thematic analysis based on sentiment, component, symptom, and solutions is adopted to fully utilize the information from reviews to enhance the performance of the probabilistic graphical model (Zheng et al., 2020, 2021). In recent work, Chakraborty and Gupta (2023) use GCN and BERT models to analyze automotive forum complaints, further validating the efficacy of data-driven techniques for recall prediction.

To further advance the defect prediction based on complaints, we take the advantage of the XGBoost model to analyze the structured complaint data in depth. The model's superiority in handling of nonlinear relationships and complex interactions in structured data enables our study to effectively capture the difference between the prediction of the first recall after launch and the subsequent recalls. Its high interpretability and robustness in processing large and noisy consumer feedback datasets offer a feasible method for effective complaint management in practice.

TABLE 1 Confusion matrix for prediction results.

Actual status	Prediction results		Total
	1 (Recalled)	0 (Non-recalled)	
1 (Recalled)	TP	FN	TP + FN
0 (Non-recalled)	FP	TN	FP + TN
Total	TP + FP	FN + TN	N

Abbreviations: FN, false negative; FP, false positive; TN, true negative; TP, true positive.

3 | CONSTRUCTION OF THE VEHICLE RECALL RISK PREDICTION MODEL

3.1 | Model evaluation metrics

Let TP (true positive) denote the number of vehicle models that are actually recalled and correctly predicted as recalled; FN (false negative) denote the number of vehicle models that are actually recalled but incorrectly predicted as non-recalled; FP (false positive) denote the number of vehicle models that are actually non-recalled but incorrectly predicted as recalled; and TN (true negative) denote the number of vehicle models that are actually non-recalled and correctly predicted as non-recalled. Based on the prediction results of the model, the confusion matrix is shown in Table 1.

The formulas that evaluate the model's classification performance are as follows (Rao et al., 2020).

Accuracy refers to the model's correct classification rate for all samples, representing the model's overall classification accuracy, that is,

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \quad (1)$$

TNR (true negative rate) refers to the model's correct classification rate for non-recalled samples, that is,

$$\text{TNR} = \frac{\text{TN}}{\text{TN} + \text{FP}} = 1 - \text{Type I error} \quad (2)$$

TPR (true positive rate) refers to the model's correct classification rate for recalled samples, that is,

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} = 1 - \text{Type II error} \quad (3)$$

AUC (area under the curve) refers to the area under the receiver operating characteristic curve (ROC curve). The ROC curve is plotted with the 1-TNR as the x-axis and the TPR as the y-axis at different threshold values. A larger AUC value indicates a stronger predictive ability of the model.

3.2 | Criteria for decision tree construction

By calculating the information gain of each feature at each threshold, the feature and its threshold that maximize the

information gain are determined for splitting the decision tree node.

- (1) Constructing the loss function. Let $l(y_j, \hat{y}_j^{(k-1)})$ be the loss function of $k - 1$ decision trees; y_j be the actual status of the j th vehicle model; $\hat{y}_j^{(k-1)}$ be the prediction result of $k - 1$ decision trees for the j th vehicle model. Adopting the logistic loss function (Zięba et al., 2016),

$$l(y_j, \hat{y}_j^{(k-1)}) = y_j \ln \left(1 + \exp \left(-\hat{y}_j^{(k-1)} \right) \right) + (1 - y_j) \ln \left(1 + \exp \left(\hat{y}_j^{(k-1)} \right) \right) \quad (4)$$

Equation (4) characterizes the error between the model's prediction results $\hat{y}_j^{(k-1)}$ and the true state y_j of the vehicle model. The smaller the value of the loss function $l(y_j, \hat{y}_j^{(k-1)})$, the smaller the error between the model's predicted values and the true values, indicating that the model's predictive performance is better.

- (2) Calculating the gradient statistics of the loss function. Let g_j be the first-order gradient statistic of the loss function $l(y_j, \hat{y}_j^{(k-1)})$, then there is (Zięba et al., 2016):

$$g_j = \frac{\partial l(y_j, \hat{y}_j^{(k-1)})}{\partial \hat{y}_j^{(k-1)}} = -y_j + \frac{1}{1 + \exp \left(-\hat{y}_j^{(k-1)} \right)} \quad (5)$$

Let h_j be the second-order gradient statistic of the loss function $l(y_j, \hat{y}_j^{(k-1)})$, then there is (Zięba et al., 2016):

$$h_j = \frac{\partial^2 l(y_j, \hat{y}_j^{(k-1)})}{\partial^2 \hat{y}_j^{(k-1)}} = \frac{\exp \left(-\hat{y}_j^{(k-1)} \right)}{\left[1 + \exp \left(-\hat{y}_j^{(k-1)} \right) \right]^2} \quad (6)$$

- (3) Measuring the information gain of features. Let x_{ij} be the value of the i th feature for the j th vehicle model; $\text{Gain}(x_{ij})$ be the information gain when the value x_{ij} of the i th feature for the j th vehicle model is used as the splitting threshold; L be the total number of vehicle models with feature values less than the threshold, which are assigned to the left branch; λ be a constant term; other parameters have the same meaning as above. Then the information gain is as follows (T. Chen & Guestrin, 2016):

$$\text{Gain}(x_{ij}) = \frac{1}{2} \left[\frac{\left(\sum_{j \in L} g_j \right)^2}{\sum_{j \in L} h_j + \lambda} + \frac{\left(\sum_{j \in R} g_j \right)^2}{\sum_{j \in R} h_j + \lambda} + \frac{\left(\sum_{j \in I} g_j \right)^2}{\sum_{j \in I} h_j + \lambda} \right] - \gamma \quad (7)$$

The larger the information gain $\text{Gain}(x_{ij})$ when the value x_{ij} of the i th feature for the j th vehicle model is used as the splitting threshold, the better the classification effect of the model.

Using the gradient statistics of the loss function of the former $K - 1$ decision trees as the information gain criterion for the K th decision tree, the splitting feature and splitting threshold of the K th decision tree node are determined based on the maximum information gain $\text{Gain}(x_{ij})$, thereby obtaining the K th decision tree.

3.3 | Construction of the decision tree

3.3.1 | Construction of the first decision tree

Step 1: Initialize the model. Before constructing the first decision tree, assign a value of 0 to the prediction results of all vehicle models $\hat{y}_j$ in the training set S (T. Chen & Guestrin, 2016).

Step 2: Determine the left and right branches. Use the value x_{ij} of the i th feature for the j th vehicle model as the splitting threshold. Vehicle models with values less than the threshold are assigned to the left branch L , while those with values greater than or equal to the threshold are assigned to the right branch R . The threshold values for each feature are the actual values from the company. Based on the above, the prediction results for both the left and right branches are $\hat{y}_j^{(0)} = 0$ at this point.

Step 3: Calculate the gradient statistics of the loss function. By substituting the true state y_j and prediction result $\hat{y}_j^{(0)} = 0$ of each vehicle model in the left branch L into Equation (5), the first-order gradient statistics of the loss function can be obtained as g_j^L . Adding them together yields $\sum_{j \in L} g_j$.

By substituting the prediction result $\hat{y}_j^{(0)} = 0$ of each vehicle model in the left branch L into Equation (6), the second-order gradient statistics of the loss function can be obtained as h_j^L . Adding up the corresponding values for all vehicle models yields $\sum_{j \in L} h_j$.

Similarly, by substituting the data of the vehicle models in the right branch R into the above process, the sum of the first-order gradient statistics $\sum_{j \in R} g_j$ and the sum of the second-order gradient statistics $\sum_{j \in R} h_j$ of the loss function can be obtained. These are two essential parameters in Equation (7).

Step 4: Determine the information gain G_1^1 for the first feature. As mentioned earlier, by substituting $\lambda, \gamma (\lambda = 1, \gamma = 1$ in this study), $\sum_{j \in L} g_j, \sum_{j \in L} h_j, \sum_{j \in R} g_j, \sum_{j \in R} h_j, \sum_{j \in I} g_j, \sum_{j \in I} h_j$ into Equation (7), the information gain G_1^1 for the first feature at the first threshold split is obtained.

Similarly, the information gain $G_1^1(u)$ for the first feature at different threshold splits can be obtained. There exists a threshold split that maximizes $G_1^1(u)$, and this maximum value is considered the information gain G_1^1 for the first feature.

Step 5: Determine the decision tree vertex. Through the process described in Steps 1–4, the information gain for each feature can be obtained as G_1^1 . There exists a feature with the maximum information gain, and the corresponding feature and threshold split are the splitting feature and splitting threshold for the decision tree vertex. Vehicle models with feature values less than the vertex splitting threshold are assigned to the left branch L , while those with values greater than or equal to the threshold are assigned to the right branch R .

Step 6: Determine the second-layer split point in the decision tree. Using the value of the first feature of the first vehicle model in the left branch L from Step 5 as the first threshold, the left branch L is further divided into left and right sub-branches.

Following the process described in Steps 3–5, the splitting feature and splitting threshold for the left sub-branch of the second layer can be obtained. Similarly, the splitting feature and splitting threshold for the right sub-branch of the second layer can also be determined. At this point, the second-layer split points of the decision tree are fully determined.

Step 7: Determine the split points in the d th layer of the decision tree and the stopping conditions. The above process only describes the construction of a two-layer decision tree. By following the process outlined in Steps 2–6, the split points of the d th layer of the decision tree can be obtained.

After each layer of the decision tree is split, the information gain of the features after splitting must be determined using Equation (7). If the information gain of a feature is less than 0, the splitting process is stopped, and the branch node becomes a leaf node.

In this study, the maximum number of layers in the decision tree is set to 10 (Bertomeu et al., 2021). When the decision tree reaches the 10th layer, regardless of whether the information gain of the features is less than 0, no further splitting is performed, and the branch nodes in the 10th layer become leaf nodes.

Step 8: Determine the output values w_t^j of the decision tree leaf nodes. Let w_t^k denote the output value of the t th leaf node of the k th feature, A denote the total number of vehicle models in the t th leaf node, and other parameters have the same meaning as mentioned above. Then (T. Chen & Guestrin, 2016):

$$w_t^k = -\frac{\sum_{j \in A} g_j}{\sum_{j \in A} h_j + \lambda} \quad (8)$$

The meaning of Equation (8): The output value of the t th leaf node where the j th vehicle model is located in the k th decision tree is the prediction result of the k th decision tree for the j th vehicle model. By substituting w_t^k into Equation (9), the recall probability of the j th vehicle model can be obtained.

Step 9: Calculate the predicted recall status for vehicle models. Let $\hat{p}_j$ denote the recall probability of the j th vehicle model, and K denote the number of decision trees. Then, the predicted probability of recall for the j th vehicle model by

the XGBoost model composed of decision trees is given by (Zięba et al., 2016):

$$\hat{p}_j = -\frac{1}{1 + \exp\left(-\left(\sum_{k=1}^K w_t^k\right)\right)} \quad (9)$$

In Equation (9), $\sum_{k=1}^K w_t^k$ is the predicted result of the j th vehicle model by the XGBoost model composed of decision trees. It is obtained by making predictions using K decision trees, resulting in w_t^k in K prediction results, and then taking the algebraic sum of these K results w_t^k . The recall discrimination threshold is set to 0.5. If the recall probability $\hat{p}_j$ of a vehicle model is less than 0.5, then the vehicle model is non-recalled, represented by 0; otherwise, it is a recalled model, represented by 1.

3.3.2 | Construction of the second decision tree

The second decision tree is constructed using the prediction results of the first decision tree. Let $k = 2$ in Equation (4), then $\hat{y}_j^{(k-1)} = \hat{y}_j^{(2-1)} = \hat{y}_j^{(1)}$.

From Equation (9), we obtain the prediction result w_t^1 of the k th ($k = 1$ at this time) decision tree for the j th vehicle model, and at this point, $\hat{y}_j^{(1)} = w_t^1$.

Substituting the true status y_j and the predicted value $\hat{y}_j^{(1)}$ into Equation (4), we obtain the loss function $l(y_j, \hat{y}_j^{(1)})$.

Substituting $l(y_j, \hat{y}_j^{(1)})$ into the process of Steps 2–8, we construct the second decision tree and obtain the output values w_t^2 of the leaf nodes of the second decision tree, which is the prediction results of the second decision tree.

3.3.3 | Construction of the k th decision tree

The gradient statistics of the loss function of the previous k decision trees are used as the information gain criterion for the k th decision tree. This is used to determine the node splits of the k th decision tree, construct the k th decision tree, and obtain the output values w_t^k of the leaf nodes of the k th decision tree.

At this point, the discriminant result of the XGBoost model composed of k decision trees is obtained by substituting the output value w_t^k of the leaf node where the j th vehicle model is located in each of the k decision trees into Equation (9). This yields the recall probability $\hat{p}_j$ of the j th vehicle model, and then comparing the $\hat{p}_j$ with the discriminant threshold of 0.5, we can obtain the predicted recall status $\hat{y}_j$ of the j th vehicle model.

3.3.4 | Determination of the maximum depth and optimal number of decision trees

Grid search and cross-validation methods are employed to determine the maximum depth and optimal number of deci-

sion trees. This process aims to optimize the performance of the XGBoost model by systematically exploring different combinations of hyperparameters. Grid search involves training and evaluating the model's performance for each combination of parameters defined in a predefined parameter grid. In this study, we set the range of decision tree depths (3–10) and the number of trees (100–800). Through grid search, we systematically calculate the validation set performance for each set of parameters, thereby determining the optimal parameter combination.

To verify the stability and generalization ability of the selected parameter combination, we adopt the k -fold cross-validation method. The dataset is divided into k equally sized subsets, with one subset left out as the validation set each time, and the remaining $k - 1$ subsets used for training. This process is repeated k times, each time selecting a different subset as the validation set, thus making full use of all the data. By comparing the average performance of different models across the k validations, we can assess the generalization ability of the model.

3.4 | Determination of vehicle model recall status

We input the vehicle model sample data, which have extracted features from the initial information that includes consumer complaint data and other information published by NHTSA, into the aforementioned XGBoost model composed of k^* decision trees to calculate the recall probability $\hat{p}_j$ of each vehicle model. If the recall probability $\hat{p}_j$ is less than or equal to 0.5, we classify the vehicle model j as a non-recalled vehicle model; otherwise, we classify it as a recalled vehicle model.

4 | EMPIRICAL ANALYSIS

4.1 | Data sources and sample selection

We identified a series of candidate variables related to defect and safety hazards based on domain-specific knowledge of automotive quality control and literature about recall risk management. Overall, the variables are complaint-, recall-, and vehicle-model-related features, such as the number of complaints, complaint sources, anomaly-related components, average mileage at failure, and historical recall records. The candidate variables were sourced from publicly available NHTSA data with sufficient completeness and accuracy. The statistical information of the dataset is shown in Table 2. NHTSA also includes recall records released by vehicle manufacturers. Sample records are shown in Table 3. Samples of consumer complaint records from the NHTSA are displayed in Table 4, showing only a portion of the data content.

By combining consumer complaint records and recall records, we construct training and test datasets for learning a recall risk prediction model for vehicle models after their initial release date. The recall risk prediction at the time of a

TABLE 2 Statistical data of National Highway Traffic Safety Administration (NHTSA).

Dataset	NHTSA
Number of consumer complaint records	1,899,972
Number of vehicle models involved in consumer complaints	16,691
Number of vehicle models involved in recall	11,400
Number of vehicle models involved in multiple recalls	2844

vehicle model's initial release uses all vehicle models in the consumer complaint records as samples, with vehicle models matched to their first recall record labeled as 1 (recalled), and vehicle models that cannot be matched to a recall record labeled as 0 (not recalled). To help predict vehicle recall events with sufficient lead time, we investigate the model's predictive performance under different time windows.

Therefore, for a specific vehicle model {Brand, Vehicle, Year}, the consumer complaint set in the datasets instances I {Brand, Vehicle, Year, Initial Release Date, Recalled or Not, Time Window, Complaint Set} for a specific vehicle model {Brand, Vehicle, Year}, it consists of consumer complaint data released during the period from {Initial Release Date} to {Initial Release Date + Time Window}. We use the **Manufacturing Start Date** of a vehicle model as its **Initial Release Date**. For data with missing Initial Release Dates, we replace them with the date of the first consumer complaint for that vehicle model.

The **Time Window** is the number of months counted after the **Initial Release Date**, used to collect the number of consumer complaints during this period. We created datasets containing consumer complaints and their corresponding recalled or not values based on a series of **Time Window** sizes (in months) = [1, 3, 6, 12, 18]. According to the processed consumer complaint dataset, nine variables were selected, and considering factors such as recalls, a total of 17 variables were chosen. The meanings of each variable are shown in Table 5. Vehicle information will be processed using feature encoding. In the data preprocessing stage of machine learning, feature encoding is a key task that involves converting nonnumerical categorical data into numerical data for algorithm processing and analysis. Label data (usually strings) are converted into an integer-encoded form, enabling algorithms to process these data internally.

For the recall risk prediction model of previously recalled vehicle models, we process the data in the same way, with the difference being that the initial release date is replaced by the recall date.

4.2 | Division of training set, validation set, and test set

To establish and validate the predictive performance of the XGBoost model, this study divides the full sample into subsets for different stages of model training and evaluation. The dataset division follows standard machine learning prac-

TABLE 3 Sample of National Highway Traffic Safety Administration (NHTSA) recall records.

Brand	Vehicle model	Year	Recall date	Manufacturing start date	Manufacturing end date
Bridgemaster V	Concrete Mixer	2013	June 12, 2013	September 9, 2011	May 8, 2013
Crane	Let2	2009	May 27, 2015	February 25, 2008	May 20, 2015
Audi	A6	2012	June 4, 2018	April 4, 2011	December 23, 2014

TABLE 4 Sample of National Highway Traffic Safety Administration (NHTSA) complaint records.

Brand	Vehicle model	Year	Complaint date	Description of specific components	Vehicle accident	Consumer's city	State	Source of complaint
Dodge	RAM 1500	2004	April 6, 2015	Power train	N	Shallotte	NC	IVOQ (NHTSA Web site)
Ford	F-150	2011	July 26, 2017	Steering	N	Hampton	GA	IVOQ
Honda	CR-V	2017	January 30, 2024	Fuel/propulsion system	N	Murphysboro	IL	IVOQ
Ford	Bronco Sport	2021	March 15, 2023	Fuel system, gasoline	N	Olympia	WA	EVOQ (hotline)

Abbreviations: Electronic Vehicle Owners Questionnaire (EVOQ, also called Hotline VOQ). IVOQ = NHTSA Web site.

tices to ensure that the model's performance can be fairly assessed on unseen data.

Before conducting the division, we first ensure the randomness of the samples to avoid any potential time series effects that may impact the model performance evaluation. Furthermore, for classification problems, especially in default prediction models, we ensure that the proportion of default and non-default samples remains consistent in each divided subset, which is achieved through stratified sampling.

Training set: 60% of the full sample is used for model training. This portion of the data is the main source for the model's learning process, where model parameters are optimized to improve the model's ability to learn from data features.

Validation set: Among the remaining 40% of the full sample, 55% is used as the validation set. The main purpose of the validation set is for model selection, including tuning the model's hyperparameters, such as the maximum depth and number of decision trees. The validation set acts as a neutral role, helping to identify overfitting phenomena on the training set and ensuring that the found parameter settings can also exhibit good generalization ability on unseen data.

Test set: Similarly, among the remaining 40% of the full sample, 45% serves as the test set. This portion of the data does not participate in the model's training and parameter selection process. The test set is used for the final evaluation of the model's performance, providing an estimate of the model's ability to predict unknown data in actual operation.

In summary, this method of dataset division ensures the efficiency of the training process while reducing the risk of overfitting and providing a reliable assessment of the model's true predictive ability. This approach ensures that when the model is finally deployed, it can make accurate and reliable predictions of default risk. The total sample sizes (N) for the recall risk prediction model at different time windows (1, 3, 6, 12, and 18 months) for a vehicle model upon its ini-

tial release are 16,691, 15,353, 13,744, 11,602, and 10,499, respectively. For the risk prediction model when the vehicle model is recalled again, the total sample sizes (N) at different time windows (1, 3, 6, 12, and 18 months) are 23,480, 21,912, 20,192, 17,594, and 15,889, respectively.

5 | CONSTRUCTION RECALL RISK PREDICTION MODEL OF VEHICLE MODELS

In this section, based on the aforementioned variables, we construct a recall risk prediction model for a vehicle model after its initial release date (Model 1) and a risk prediction model for a recurring recalled vehicle model (Model 2). By processing the training data through the decision tree construction process described above, an XGBoost model consisting of k ($k = 1, 2, \dots, 100$) decision trees can be established. When the time window is set to 1 month, the 70th decision tree in the XGBoost model for constructing Model 1 is shown in Figure 1.

The values of important parameters are adjusted using a combination of hyperparameter grid search and cross-validation methods. Similarly, XGBoost Model 1 and Model 2 with different time windows can be constructed. The results of the adjusted important parameters are shown in Table 6.

After completing the parameter tuning, the optimal models are obtained. The generalization ability of the models is tested using the test set samples. The area under the ROC curve is used to evaluate the prediction accuracy of the vehicle recall risk prediction models under different time windows, as shown in Figures 2 and 3. The AUC values of Model 1 are 0.97, 0.94, 0.86, 0.81, and 0.76, respectively, while the AUC values of Model 2 are 0.82, 0.82, 0.82, 0.81, and 0.84, respectively, indicating that both models possess good predictive capabilities.

TABLE 5 Meaning description of variables.

Category	Name	Meaning
Consumer complaint set	Total number of complaints	Total number of complaints of a vehicle model in the consumer complaints records
	Number of complaints involving accidents	Number of complaints involving traffic accidents of a vehicle model in the consumer complaints records
	Number of complaints involving injuries	Number of complaints involving consumer injuries of a vehicle model in the consumer complaints records
	Number of complaints involving deaths	Number of complaints involving consumer deaths of a vehicle model in the consumer complaints records
	Number of sources of complaints	The number of consumer complaint sources involved in the complaint set of a vehicle model. Complaint sources include consumer action groups; referrals from congressional offices; defect petitions and their results; hotline VOQs; early warning reports; insurance companies; the National Highway Traffic Safety Administration (NHTSA) website; consumer letters; the NHTSA mobile app; optical character recognition (OCR) VOQs; recall complaints and the results of recall investigations; recall petitions and their results; portable safety complaint forms (PDF); and NHTSA vehicle owner questionnaires.
	Average mileage at failure	The average failure mileage of a vehicle model in the consumer complaints records
	Variance of complaint cities	Vehicle model is involved in different cities in the consumer complaints records, and its variance is calculated.
	Variance of complained components	Vehicle model is involved in different components in the consumer complaints records, and its variance is calculated.
Historical recalls	Number of complained components	The vehicle model is involved in different components in the consumer complaints records, and the total number of components complained is calculated.
	Number of brand's recalls in history	Number of recalls the brand has been involved in historically
	Total number of potential consumers affected by brand historical recalls	Brand's total number of potentially affected consumers involved in historical recalls
	The average number of potential consumers affected by brand historical recalls	Brand's average number of potentially affected consumers per recall in its history
	Variance of brand's history recall components	The number of components involved in the brand's historical recalls, and the variance of the recalled components is calculated.
	Number of recalled components in brand historical recalls	The components involved in the brand's historical recalls and the total number of recalled components
	Number of ODI recalls in brand historical recalls	Number of recalls the brand has been involved in historically, initiated by ODI
	Number of MFR recalls in brand historical recalls	Number of recalls the brand has been involved in historically, initiated by MFR
Vehicle	Number of OVSC recalls in brand historical recalls	Number of recalls the brand has been involved in historically, initiated by OVSC
	Brand	Brand, feature encoded
	Vehicle model	Vehicle model, feature encoded

Abbreviations: ODI, Office of Defects Investigation; MFR, Manufacturer Recall; VOQ, Vehicle Owners Questionnaire; OVSC, Office of Vehicle Safety Compliance.

6 | INTERPRETING RECALL RISK PREDICTION USING SHAP FEATURE ANALYSIS

Although the XGBoost algorithm demonstrates excellent predictive performance, its model interpretability is relatively poor, especially in explaining the mechanism of how various variables jointly influence the prediction results. To

compensate for this shortcoming, SHapley Additive exPlanations (SHAP) values are introduced to improve the model's interpretability. Through SHAP values, we can quantify the contributions of features to the model's predictions.

SHAP values provide an intuitive visual representation, where the position of each data point reflects its feature's contribution to the model's prediction. As shown in Figure 4, features are ranked by their importance on the vertical axis,

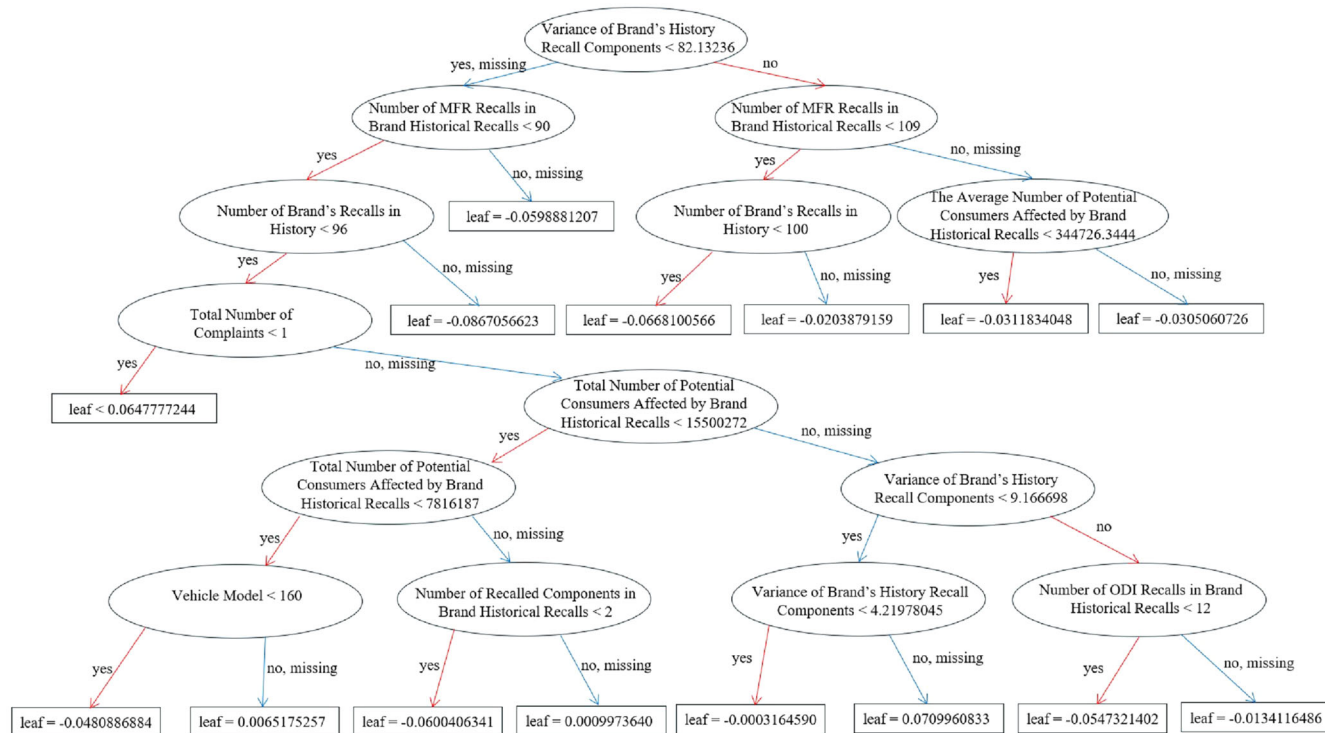


FIGURE 1 Decision tree (70th tree) in the XGBoost model with a 1-month time window for Model 1.

TABLE 6 Best parameters of XGBoost models with different time windows.

Model	Time window (month)	Number of decision trees	Learning rate	Maximum tree depth	Training sample ratio	Sampling rate
Model1	1	100	0.1	7	0.8	0.8
	3	200	0.1	5	0.8	1
	6	200	0.1	5	0.8	1
	12	200	0.1	7	0.8	0.8
	18	200	0.1	7	1	0.8
Model2	1	400	0.1	7	0.8	1
	3	400	0.1	7	0.8	1
	6	400	0.1	7	0.8	0.8
	12	400	0.1	7	0.8	0.8
	18	400	0.1	7	0.8	1

with the most influential ones at the top. The horizontal axis shows the magnitude of the SHAP values. Positive SHAP values mean the features increases the likelihood of the predicted outcome, while negative values mean the opposites. Red represents high feature values, and blue low values.

For a given feature, its distribution pattern indicates the level of uncertainty in the model output. A concentrated pattern suggests a consistent impact, while a scattered one indicates greater variability. The visualization makes SHAP values an effective tool for interpreting the prediction of complex machine learning models.

The SHAP feature summary plot for model (Figure 4) shows that for newly launched vehicle models, the impact of complaint numbers on the risk of recall varies, depending on factors like the severity and complaint channels. A small number of complaint sources hint at a high recall risk. Similarly, an increase in the average number of users affected by historical recalls of a brand signals greater recall risk, implying potential design or manufacturing defects. Low values for the number of components involved in complaints are associated with a high recall risk, suggesting that quality issues in a few core components can trigger recalls. Additionally, lower variance in the components involved in a brand's

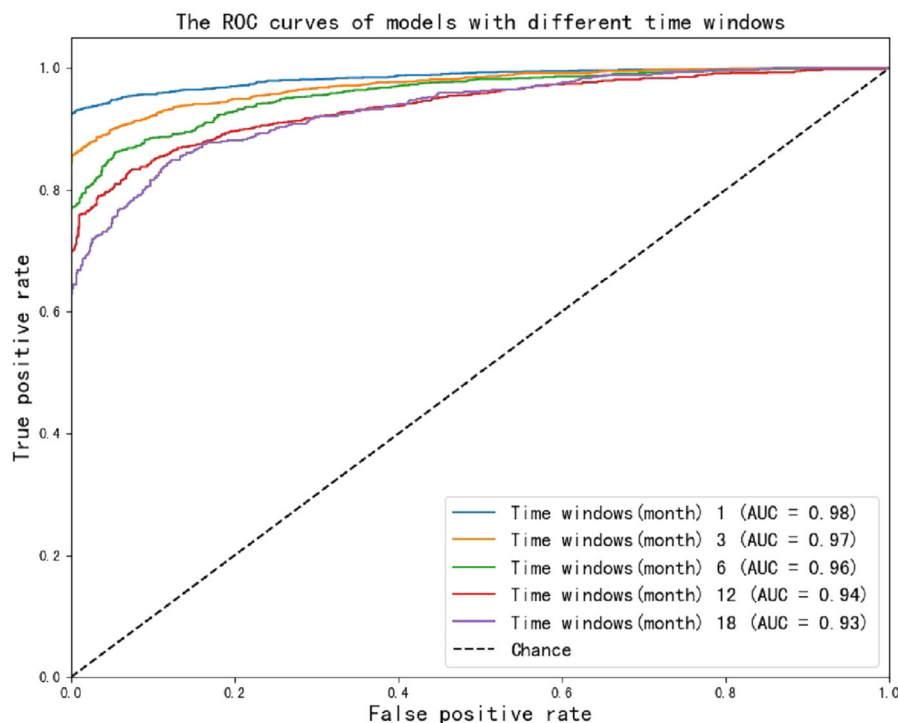


FIGURE 2 Receiver operating characteristic (ROC) curves of Model 1. AUC, area under the curve.

historical recall events is related to a high recall risk, possibly indicating persistent quality issues in specific components.

Factors such as the average mileage at failure, brand, and vehicle model also play significant roles in predicting the recall risk. The link between component durability and the overall vehicle reliability makes the average mileage at failure a crucial indicator. Brand and vehicle model can reflect underlying attributes like manufacturing quality, corporate reputation, and safety standards. Some brands may have lower recall risks due to strong quality control and extensive manufacturing experience, while others may face higher recall risks due to weak oversight. Vehicle model characteristics, including design complexity, material choice, and component quality, also influence recall likelihood.

Figure 5 presents the SHAP analysis for Model 2, providing insights into the risk of recurrent recalls. The analysis shows that low variance of components in a brand's historical recalls is linked to a higher recall risk, suggesting persistent quality issues in specific components. Fundamentally unresolved issues may increase the change of another recall.

The SHAP value distribution for brand and vehicle models follow a complex pattern, which reflects the diversity of brands and models in predicting the risk of recurrent recalls. A brand with past recalls involving many components may indicate widespread quality control issues or design flaws, increasing the risk of future recalls. A large number of complaints may signal ongoing or unresolved problems, further increasing recall likelihood.

Comparing the SHAP results for predicting the risk of the initial recall and the subsequent recalls reveals shifts in how

different features influence recall risk. Initial recalls may be driven by new model issues, while subsequent recalls are affected by persistent quality concerns. These insights help automakers develop data-driven recall management strategies to address quality and safety challenges effectively.

7 | MODEL PERFORMANCE EVALUATION

To validate the performance of the XGBoost model in predicting vehicle recall risk, we compared it with traditional machine learning algorithms such as logistic regression, SVMs, and random forests. The evaluation metrics include accuracy, recall, and AUC value, which cover multiple aspects of the models' overall predictive performance. Table 7 presents the performance comparison of these models across different time windows.

In the recall risk prediction model for the initial recall (Model 1), XGBoost's performance gradually declines as the time window extends, reflecting the challenge of using historical data to predict future risks over time. However, even with an 18 months prediction span, XGBoost maintains high performance. Compared to other models, XGBoost consistently outperforms them across all time windows.

In the recall risk prediction model for subsequent recalls (Model 2), XGBoost maintains its lead in accuracy and AUC values across various time windows, demonstrating its superiority in predicting the risk of subsequent recalls. While other algorithms remain relatively stable, their lower recall rates indicate challenges in identifying actual recall events.

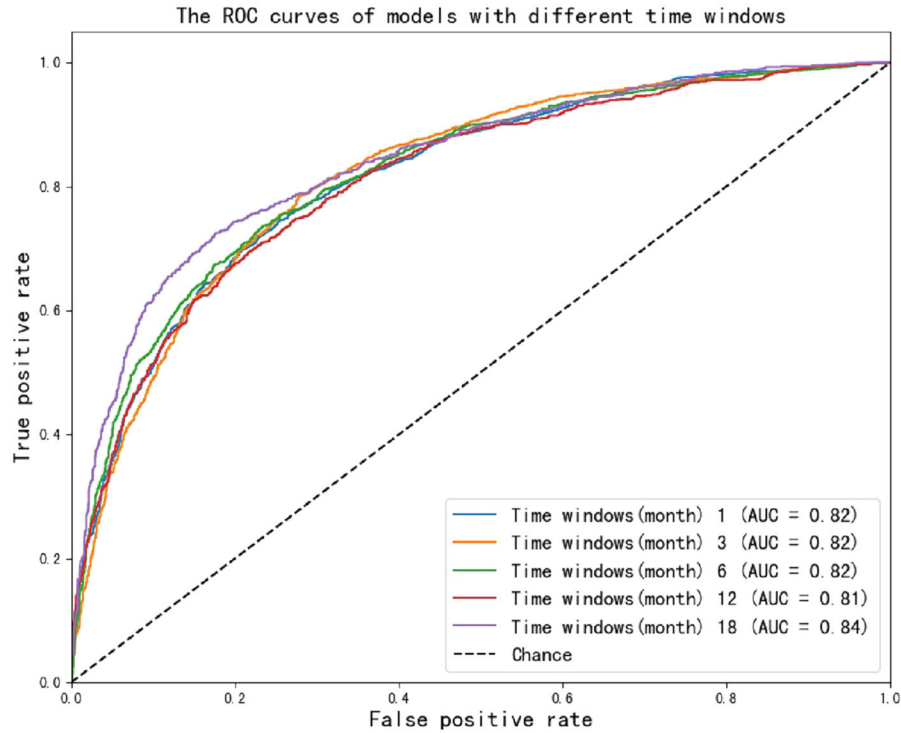


FIGURE 3 Receiver operating characteristic (ROC) curves of Model 2. AUC, area under the curve.

TABLE 7 Comparison of model results.

Time window (month)	Model	Model 1			Model 2		
		Accuracy	Recall	AUC	Accuracy	Recall	AUC
1	XGBoost	0.95	0.93	0.98	0.74	0.73	0.82
	Random forest	0.95	0.92	0.98	0.66	0.79	0.70
	Logistic regression	0.94	0.93	0.97	0.57	0.58	0.60
	SVM	0.95	0.92	0.97	0.60	0.63	0.63
3	XGBoost	0.91	0.88	0.97	0.75	0.73	0.82
	Random forest	0.90	0.84	0.95	0.63	0.50	0.69
	Logistic regression	0.88	0.88	0.92	0.58	0.43	0.59
	SVM	0.90	0.86	0.95	0.59	0.46	0.62
6	XGBoost	0.89	0.85	0.96	0.75	0.72	0.82
	Random forest	0.86	0.77	0.94	0.63	0.34	0.67
	Logistic regression	0.80	0.84	0.88	0.60	0.33	0.60
	SVM	0.87	0.80	0.92	0.61	0.32	0.62
12	XGBoost	0.87	0.81	0.94	0.76	0.66	0.81
	Random forest	0.83	0.69	0.92	0.68	0.22	0.69
	Logistic regression	0.76	0.78	0.84	0.66	0.20	0.61
	SVM	0.83	0.73	0.89	0.67	0.18	0.64
18	XGBoost	0.86	0.80	0.93	0.81	0.67	0.84
	Random forest	0.82	0.63	0.91	0.73	0.15	0.71
	Logistic regression	0.74	0.74	0.81	0.72	0.14	0.63
	SVM	0.82	0.68	0.87	0.72	0.11	0.66

Abbreviations: AUC, area under the curve; SVM, support vector machine.
The bolded numbers indicate the maximum values across the given aspects under different time windows.

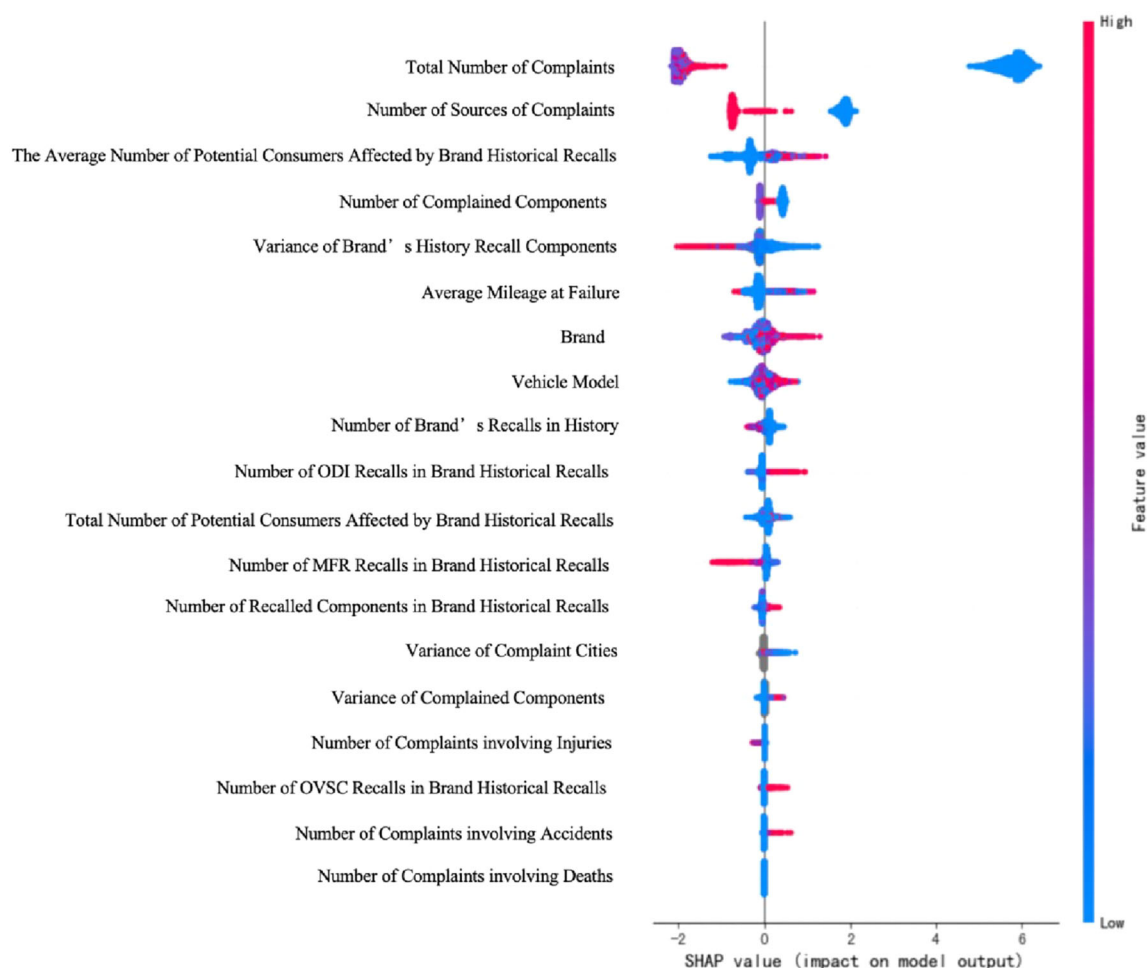


FIGURE 4 SHapley Additive exPlanations (SHAP) feature analysis of Model 1.

Overall, XGBoost exhibits exceptional performance in both prediction models, particularly in accuracy and AUC values, reinforcing its applicability for recall risk prediction. Although all models experience performance declines over longer prediction periods, XGBoost remains the most reliable option. The low recall rates of random forest models may stem from their insufficient sensitivity to imbalanced data. Logistic regression and SVM, while maintaining relatively stable accuracy, show limitations in recall and AUC values, suggesting that they may be better at predicting non-recall events but less effective in identifying true recalls. Therefore, XGBoost proves highly effective for recall risk prediction, especially within shorter time windows.

The model's performance gradually declined as the time window expanded, which suggests the difficulty of predicting risks using long-term historical data. We find that short-term windows (e.g., 1–3 months) provide clearer signals. During this period, consumer complaints often reflect early-stage design or manufacturing defects, appearing in concentrated and intense patterns and resulting in high-quality, low-noise input data. As the time window expands to 6, 12, or even 18 months, accumulated complaints introduce noise generated by product improvements, market feedback, or consumer ran-

dom events, weakening the correlation between complaints and product defects. While long-window data may show smoother or more heterogeneous distributions, it impairs both the model's interpretability and predictive power when predicting defects. Furthermore, consumers' likelihood of complaint and their adoption of complaint sources may change in the life cycle of an automotive model, leaning to the changing weights of predictors during different time windows. The decline requires more advanced algorithms or the input of additional variables to enhance the accuracy of predictions.

8 | CONCLUSION

This study employed the XGBoost model to conduct an in-depth analysis of the key risk factors associated with vehicle recalls in the US market, providing valuable proactive risk management support for automakers and regulatory agencies. We leveraged the extensive data resources from NHTSA to construct high-precision recall risk prediction models. The models exhibited exceptional performance across different time windows, particularly maintaining a high level of AUC values over a prediction timespan of up to 18 months, demon-

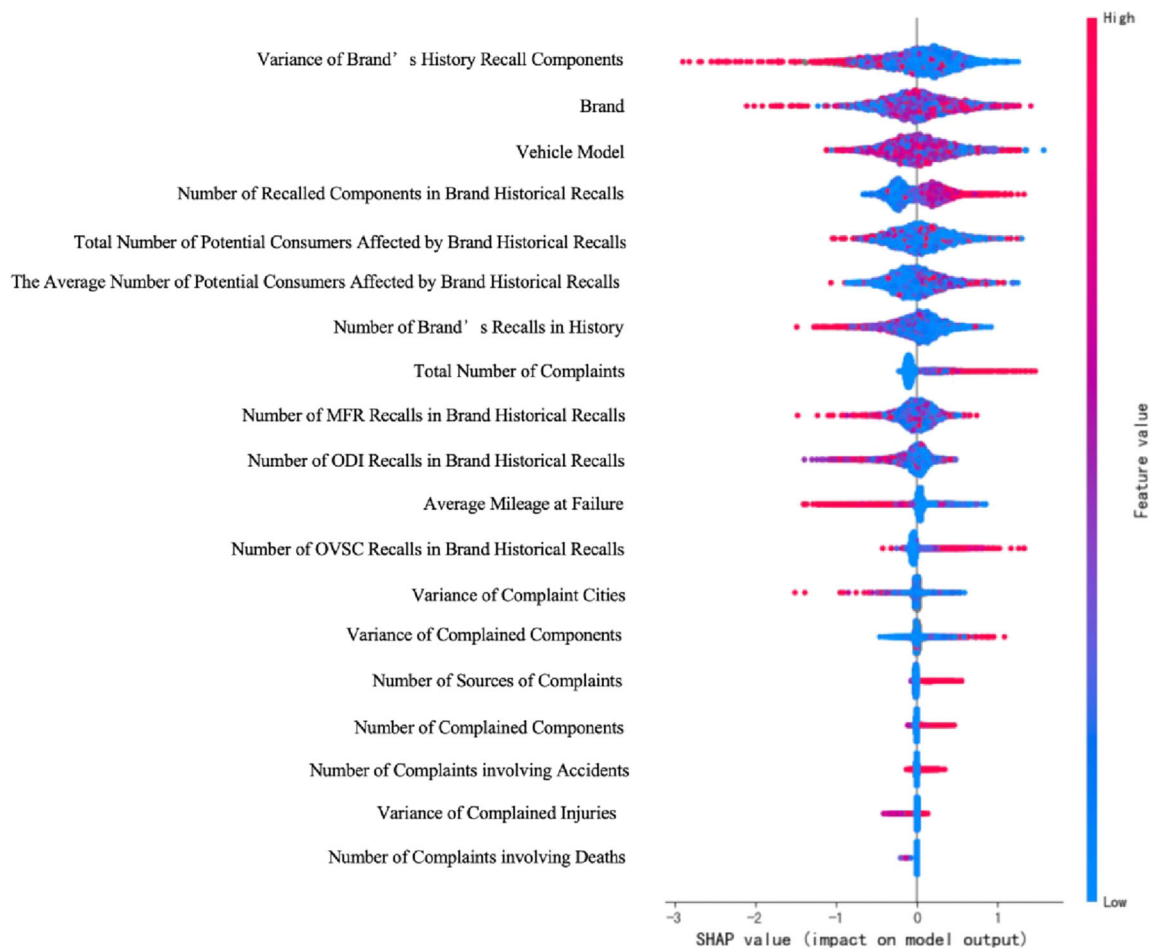


FIGURE 5 SHapley Additive exPlanations (SHAP) feature analysis of Model 2.

strating their predictive accuracy and stability, and therefore offering an approach to effectively engage consumers' negative feedback in product quality management.

The study revealed that using short-term window data can significantly enhance prediction accuracy, which is crucial for developing responsive and timely management strategies and resource allocation. The research also found that data within short-term windows can more accurately capture imminent risks, providing key information for swift responses to market changes and regulatory requirements. Moreover, the introduction of SHAP values substantially improved the interpretability of the XGBoost model, revealing decisive factors of model predictive performance, such as the number and source of complaints and the frequency of historical recalls by brand. These in-depth analyses of key indicators greatly enhanced the effectiveness of the prediction models, opening up new avenues for improving predictive accuracy.

The key innovations of this study lie first in the differentiation between the recall risk of newly launched vehicle models and the risk of repeated recalls for vehicle models with a recall history. By precisely analyzing the characteristics of these two distinct stages of vehicle models, two independent and efficient prediction models were constructed. This

approach not only improved the accuracy of predictions but also provided automakers with more scientific quality control and recall strategies. Additionally, by extracting and analyzing key features influencing recall risk, such as quantitative indicators of consumer complaint data and historical recall data by brand, this step not only enhanced the practical application value of the prediction models but also greatly improved model interpretability, enabling manufacturers and regulatory agencies to more accurately understand and respond to recall risks. Finally, by comparing the effects of prediction models under different time windows, the study found that data within short-term windows have more precise predictive power for immediate risks. This discovery guides manufacturers to be more responsive to market and regulatory demands in terms of risk management and resource allocation.

9 | LIMITATIONS AND FUTURE WORKS

Automobile manufacturers can adopt our predictive model in their quality control to detect vehicle recall risks effectively and take proactive measures before the risk escalates.

However, its practical implementation demands overcoming challenges in data integration, model adaptations, and decision implementation. First, manufacturers need to integrate accurate and complete information on internal production, maintenance, and quality inspection data from different systems into a unified platform and adhere to regulatory standards to use the privacy-related complaint information. Second, to maintain the model predictive power that may be reduced due to data lag and noise over time, manufacturers must implement regular model retraining and online learning. Third, bridging technical insights from SHAP values to actionable decisions requires collaboration between data teams and stakeholders and the balance between prevention costs against risk reduction benefits. Addressing these challenges ensures the model's reliability and alignment with operational realities.

Although the XGBoost model has demonstrated outstanding efficiency and accuracy in predicting automotive recall risks in the US market, its performance depends on the quality and completeness of historical data (T. Chen & Guestrin, 2016). Therefore, future research could explore models that incorporate more heterogeneous data sources to enhance the accuracy and robustness of predictions.

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ORCID

Jiuchang Wei  <https://orcid.org/0000-0003-4884-0669>

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